

TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

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ABSTRACT

This paper studies a failure mode and a remedy for model merging in the low-rank regime. Model merging is typically framed as compressing several fine-tuned models into one set of weights, but this framing is brittle when the source models are already compressed through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). Our first contribution is diagnostic: we show that conventional merging methods applied directly to low-rank weights can severely degrade the merged model. Our second contribution is a different merging objective: rather than collapse all adapters into one model, we keep a compact basis, equivalent to storing two or more components, from which each task-specific model can be reconstructed by linear combination. This turns merging into the construction of a reconstruction-capable model space and makes it possible to “revert” to the specialized model needed for a task. Our third contribution is Reversible Model Merging (RMM), an efficient, data-free method with a closed-form solution for the basis weights and task-specific coefficients. Across datasets, compression ranks, and model scales, RMM preserves low-rank model performance substantially better than existing merging baselines while exposing an explicit storage–accuracy trade-off.

1 INTRODUCTION

Modern adaptation pipelines increasingly combine two efficiency tools: parameter-efficient fine-tuning and model merging. Low-Rank Adapters (LoRA) (Hu et al., 2021) adapt large language models without full retraining, while post-training singular value decomposition (SVD) truncation reduces storage and inference costs after fine-tuning (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Model merging addresses a complementary need: consolidating multiple task-specific models into a shared deployment artifact. Existing methods, from parameter interpolation to optimization-based merging, show that a merged model can be competitive across source tasks and can often be produced without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

The central limitation we surface is that the usual single-model target can hide substantial task-level degradation. Even strong merging strategies may not match the original fine-tuned models, and the gap becomes especially visible when the models to be merged are low-rank LoRA or PT-SVD representations. The reason is structural: a single merged parameter vector must collapse distinct task-specific directions into one space, so the specialized adaptations that make each model effective can interfere with one another (Gargiulo et al., 2025).

This failure matters because low-rank adaptation is often used precisely in settings where many task-specific models accumulate. Multi-task, federated, and continual learning systems may maintain dozens or hundreds of low-rank adapters, each inexpensive alone but costly to store, route, and manage at scale. Keeping every adapter weakens the storage advantage of low-rank parameterization, whereas collapsing them into one adapter sacrifices task performance. The paper therefore targets a reviewer-relevant gap: how to reduce redundancy among low-rank task models without erasing the ability to recover the specialized models that originally performed well.

Our first contribution is an empirical and conceptual diagnosis of this gap. We study deltas, the weight differences between each fine-tuned model and the shared pre-trained initialization, and ask whether existing merging techniques remain effective after those deltas are represented in low-rank

form. They do not: directly merging low-rank deltas produces catastrophic degradation. The evidence points to two causes that are specific to the compressed setting: low-rank updates have limited expressive capacity, and independently optimized task subspaces can be misaligned, so merging them amplifies interference rather than preserving task structure.

Our second contribution is to change the object being optimized. Instead of asking how to force all low-rank adapters into one final model, we ask whether merging can keep enough structure to reconstruct the original adapters on demand. RMM answers this by storing a compact set of weights as a shared basis and recovering each task model through a linear combination of that basis. The contribution is not a new task model or a new dataset; it is a revised merging formulation that preserves complementary directions across a small set of basis components and supports post-merging “reversion” to the task-specific adapters.

Our third contribution is the resulting method, Reversible Model Merging (RMM). We formalize low-rank merging as a basis-selection problem whose objective is reconstruction error, derive a closed-form solution for the optimal basis and task-specific coefficients, and implement the procedure without training data. We then validate the method across datasets, compression ranks, and model configurations. The experiments show that RMM consistently outperforms conventional merging baselines for low-rank models, while its basis size provides a direct knob for trading storage against reconstruction fidelity. In summary, the paper contributes (i) a failure analysis for conventional low-rank merging, (ii) a reversible reconstruction objective, and (iii) a closed-form, data-free method that makes the storage–performance trade-off explicit.

2 RELATED WORK

We position RMM against existing merging work by separating what prior methods contribute from what they leave unresolved in the low-rank setting. Most methods aim to produce one merged model, either through data-free parameter-space operations or through optimization with data or auxiliary objectives. This section reviews those families because they form the baselines and also clarify the paper’s distinction: RMM keeps the data-free appeal of parameter-space merging but changes the target from one aggregate model to a reversible basis.

Task Arithmetic: Task arithmetic (Ilharco et al., 2023) is the basic data-free reference point for our study. It represents each task by a delta between the fine-tuned weights and the pre-trained initialization, then composes those deltas by addition, subtraction, or scaling. Its contribution is the observation that task behavior can often be manipulated directly in weight space. Our experiments ask whether this linear compositionality still holds when the deltas being composed are low-rank factors rather than full task updates.

DARE: DARE contributes a pruning-based response to interference in task deltas (Yu et al., 2024). Rather than merge every parameter change, it randomly drops a large fraction of updates—up to 90-99%—and rescales the remaining entries to preserve update magnitude. This design reduces some conflicts in full-parameter merging, but it still ends in a single merged representation. RMM differs by treating interference in low-rank models as evidence that reconstruction, not one-shot aggregation, should be the merging objective.

TIES: TIES-merging contributes a deterministic alternative for conflict resolution (Yadav et al., 2023). Its Trim, Elect Sign, and Merge stages keep large-magnitude updates, enforce sign agreement, and average the surviving parameters. The resulting consensus model is more stable than naive averaging, and related work such as EMR-Merging further reduces interference by electing a unified model and attaching task-specific binary masks and rescalers (Huang et al., 2024). These methods are important comparisons because they explicitly address interference, but they still assume that the end product is a unified merged model rather than a compact representation from which task models can be recovered.

Other Merging Methods: Optimization-based merging contributes stronger control over the final model, usually by spending data or computation. Model soups (Wortsman et al., 2022) tune interpolation weights on validation loss, and Fisher-weighted averaging (Matena & Raffel, 2022) uses second-order estimates to prioritize parameters important for task performance. RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) search for task-dependent coefficients using objectives based on prediction discrepancy or unlabeled prediction entropy, and recent work uses a first-order

Taylor surrogate for the loss difference between merged and individual models (Wei et al., 2025). These methods motivate the importance of principled objectives, but their reliance on labeled or unlabeled data, extra optimization, or both makes them less aligned with the data-free low-rank setting targeted here.

3 PRELIMINARIES

We first state the standard merging problem so that the departure introduced by RMM is explicit. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. Conventional merging searches for an algorithm $\mathcal{M}$ that maps these models to one set of weights, $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. Under this framing, the desired procedure is the one that maximizes average validation performance across all tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

Here $\mathcal{D}_i$ is the held-out validation set for task i , and Perf can be any task-appropriate metric, such as accuracy or F1 score. We use the standard assumption that all fine-tuned models share the same pretrained initialization θ_{pre} . For layer l , the task-specific update is the delta $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, with $\Delta_i^l \in \mathbb{R}^{m \times d}$ for a layer of size $m \times d$. This delta notation is the bridge between prior single-model merging and our reconstruction-based formulation.

3.1 LOW-RANK COMPRESSED MODELS

The paper’s specific setting is merging after the fine-tuned updates have been compressed into low-rank form. This assumption is practically important because it is the source of the storage benefit and also the source of the merging difficulty. We consider two common ways to obtain such low-rank updates:

LoRA fine-tuning: LoRA contributes a low-rank training path: each layer receives trainable matrices $A \in \mathbb{R}^{m \times r}$ and $B \in \mathbb{R}^{r \times d}$, where $r \ll \min(m, d)$. The pretrained weights remain frozen, so only the low-rank factors A and B are optimized.

Post-training SVD truncation (PT-SVD): PT-SVD contributes a post hoc compression path: after standard fine-tuning, the update Δ_i^l is decomposed by SVD and truncated to the top- r singular components. The resulting approximation is $\hat{\Delta}_i^l = A_i^l B_i^l$, with $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$ for $r \ll \min(m, d)$.

Both compression paths lead to the same mathematical object for our purposes: the layer- l update for task model i is stored as $\hat{\Delta}_i^l = A_i^l B_i^l$, where $A_i^l \in \mathbb{R}^{m \times r}$, $B_i^l \in \mathbb{R}^{r \times d}$, and r is the compression rank. Conventional low-rank merging would therefore try to combine the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ into one model:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where L is the number of layers. This equation captures the baseline target that RMM revises: instead of only asking for one merged low-rank model, we ask whether a compact representation can retain enough information to reconstruct the task-specific low-rank deltas. The next section shows why this change is necessary.

4 MOTIVATING OBSERVATION

The motivating contribution of this section is a controlled stress test of the standard merging objective under low-rank compression. For each layer l , prior work suggests two natural ways to extend conventional merging to low-rank factors (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

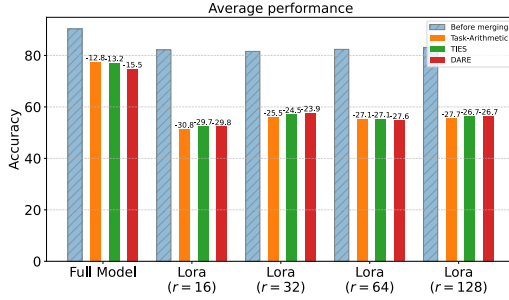


Figure 1: Motivating failure case for LoRA merging: average task performance of individual models before merging compared with conventional merged models, for full fine-tuning and LoRA ranks $r \in \{16, 32, 64, 128\}$.

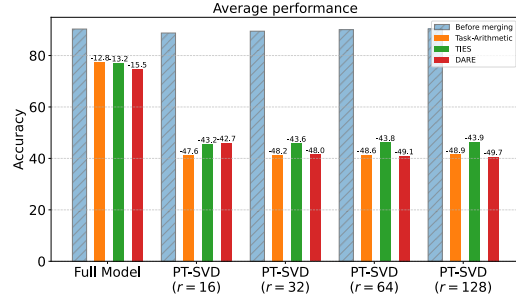


Figure 2: Motivating failure case for PT-SVD merging: average task performance of individual models before merging compared with conventional merged models, for full fine-tuning and PT-SVD ranks $r \in \{16, 32, 64, 128\}$.

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l \mathbf{B}_i^l\}_{i=1}^n),$$

where each low-rank update is first reconstructed into its full-rank form before merged.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l\}_{i=1}^n) \mathcal{M}(\{\mathbf{B}_i^l\}_{i=1}^n),$$

where the low-rank factors are merged independently before recomposition.

Both extensions are mathematically plausible, but only the separate version preserves the efficiency motivation for low-rank models. Combined merging must materialize and store an $m \times d$ update, whereas separate merging keeps the low-rank scale of $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$. Because $r(m + d) \ll md$ when $r \ll \min(m, d)$, combined merging turns a compressed representation back into a full update. For example, with $m = d = 1024$ and $r = 16$, combined merging stores $md = 1,048,576$ parameters, while separate merging stores $r(m + d) = 16 \times (1024 + 1024) = 32,768$ parameters.

This **32 $\times$ storage reduction** makes separate merging the practical baseline for compressed models. We therefore evaluate conventional methods in the setting that is most favorable to low-rank efficiency, rather than attributing failure to an obviously storage-inefficient implementation.

Using this separate strategy, we apply existing merging baselines to low-rank compressed RoBERTa-base models (Zhuang et al., 2021) across five GLUE tasks (Wang et al., 2018); Appendix A.3 gives the full tables. Figures 1 and 2 show the key finding: once the source models are low-rank, conventional merging causes a *catastrophic drop* in average task performance relative to the individual models and to full-rank merging. This diagnostic result supports two mechanisms: (i) **Amplified task interference:** compression removes representational capacity, so low-rank models have less slack for absorbing destructive interactions; (ii) **Loss of coupling between A and B:** conventional separate merging treats $\mathbf{A}_i^l$ and $\mathbf{B}_i^l$ independently, even though their product defines the update, which can misalign the reconstructed deltas. Together, these mechanisms explain why standard merging is poorly matched to compressed representations.

The conclusion of the motivating study is the design premise of the paper: for low-rank weights, a single merged model is not a reliable carrier of all task-specific behavior. We therefore ask a different question: *Can we approximate the original compressed models by storing a compact set of basis models, with the budget corresponding to two or more adapters?* This question leads directly to our main method section, where merging is formalized as building a *reconstruction-capable model space* rather than a single collapsed representation.

5 REVERSIBLE MODEL MERGING

RMM is the paper’s methodological contribution: it replaces the single merged model with a compact, reversible representation. The motivating experiments show that one set of weights cannot

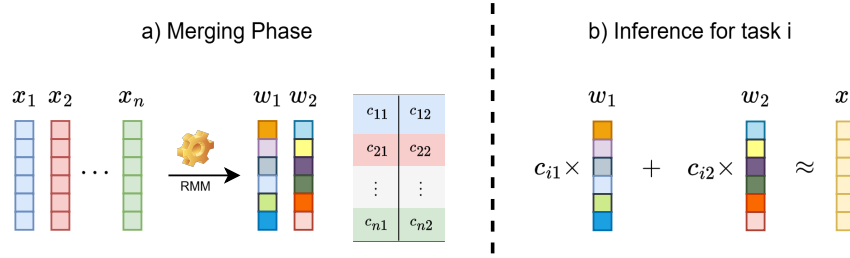


Figure 3: RMM as a reversible contribution to low-rank merging. **Left:** The offline merging phase stores a compact basis (two directions in this example) plus task-specific coefficients. **Right:** At inference, the requested task model i is reconstructed by the corresponding linear combination.

faithfully preserve all low-rank task behaviors. RMM therefore stores a small set of shared basis components from which the original task models can be approximately reconstructed.

The key design choice is to retain p merged components, with $2 \leq p \ll n$, instead of forcing all n task-specific models into a single solution $\hat{\theta}_{\mathcal{M}}$. These components span a low-dimensional subspace intended to capture the dominant variation across tasks. The basis size p is therefore a visible contribution of the method: it gives practitioners a direct way to trade additional storage for better reconstruction accuracy.

We make the reversible representation local to the low-rank factors. For layer l , the relevant r -dimensional objects are the rows of A_i^l and the columns of B_i^l , where r is the compression rank. We call each such row or column a *task vector*, and a fixed row or column index a *task vector position*. At a fixed position, the vector from model i is denoted $x_i \in \mathbb{R}^r$, and RMM compresses the collection $\{x_i\}_{i=1}^n$ into a shared basis for later reconstruction. Because there are m row positions in A_i^l and d column positions in B_i^l , the same basis-selection procedure is applied independently to all $m + d$ positions in each layer.

At each task vector position, RMM stores a basis $\{w_1, \dots, w_p\}$ with $w_j \in \mathbb{R}^r$ for $1 \leq j \leq p$. The matrix $W = [w_1 \dots w_p] \in \mathbb{R}^{r \times p}$ defines the reconstruction subspace, and the coefficient matrix $C \in \mathbb{R}^{n \times p}$ stores one row $[c_{i1}, \dots, c_{ip}]$ per task model. The method’s objective is therefore explicit and reviewer-facing: choose the shared basis and the task-specific coefficients that minimize total reconstruction error across the n task vectors:

$$\min_{W, C} \sum_{i=1}^n \left\| x_i - \sum_{j=1}^p c_{ij} w_j \right\|_2^2. \quad (3)$$

This formulation makes the storage contribution transparent. Instead of retaining all rn task-vector parameters, RMM stores rp basis parameters and np coefficient parameters, for a total of $p(r + n)$. The tunable value of p controls the storage–accuracy trade-off, and Figure 3 illustrates the case $p = 2$. The optimization in equation 3 can be written in matrix form as:

$$\min_{W, C} \|X - CW^\top\|_F^2, \quad (4)$$

Here $X = [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ stacks all task vectors for the current position. For any fixed basis W , the optimal coefficients are given by the least-squares solution $C^* = XW(W^\top W)^{-1}$. With an orthonormal basis, $W^\top W = I_p$ and $p < r$, the contribution becomes a standard subspace-reconstruction problem:

$$\min_W \|X - XWW^\top\|_F^2 \quad \text{s.t.} \quad W^\top W = I_p. \quad (5)$$

Operationally, this means RMM stores the basis $W \in \mathbb{R}^{r \times p}$ and the projected coefficients $XW \in \mathbb{R}^{n \times p}$, where XW corresponds to C in equation 4.

Theorem 1 (Optimal Solution to equation 5). *Let $X \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis W^* that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $X^\top X$. Equivalently, W^* consists of the top- p right singular vectors of X .*

The proof of Theorem 1 appears in Appendix A.1. The theorem supplies the closed-form core of RMM, and the implementation separates naturally into an offline merging phase and an inference-time reconstruction phase.

5.1 MERGING PHASE

The offline merging phase builds the compact representation that replaces storing every task vector. For each layer and each task vector position, we stack the vectors from all n models as $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$. We compute $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and center the matrix as $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$. We then take the SVD $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ and keep the top- p right singular vectors, $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$, which Theorem 1 identifies as the optimal orthonormal reconstruction basis for equation 5. Finally, we store $\mathbf{C}^* = \mathbf{X}\mathbf{W}^* \in \mathbb{R}^{n \times p}$; its i -th row $[c_{i1}, \dots, c_{ip}]$ is the coefficient vector for reconstructing task model i .

After this phase, the original task vectors $\{\mathbf{x}_i\}_{i=1}^n$ are no longer stored. Each position keeps only $\mathbf{W}^*$, $\mathbf{C}^*$, and $\boldsymbol{\mu}$, reducing storage from rn parameters to $p(r + n) + r$ parameters per task vector position. This is an offline procedure, so the additional computation is paid before inference rather than during task execution.

5.2 INFERENCE PHASE

At inference time, RMM contributes reversibility rather than another universal model. As in common model-merging evaluations, we assume the target task is either specified by the user or selected by an oracle router. Given task index i , RMM reconstructs each discarded task vector as $\hat{\mathbf{x}}_i$ from the stored tuple $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$. Concatenating the reconstructed vectors across positions recovers the rows of $\mathbf{A}_i^l$ and the columns of $\mathbf{B}_i^l$.

For each position, the stored objects have clear roles: $\mathbf{W}^* \in \mathbb{R}^{r \times p}$ contains the shared orthonormal basis, $\mathbf{C}^* \in \mathbb{R}^{n \times p}$ contains task-specific coefficients, and $\boldsymbol{\mu} \in \mathbb{R}^r$ restores the mean. The reconstruction is the corresponding coefficient-weighted basis combination plus the mean:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The vector $\hat{\mathbf{x}}_i \in \mathbb{R}^r$ approximates the original $\mathbf{x}_i$ removed during merging. Applying the same reconstruction independently to every row of $\mathbf{A}_i^l$, every column of $\mathbf{B}_i^l$, and every layer yields the reconstructed low-rank delta $\hat{\Delta}_i$ for task model i . Algorithm 1 summarizes the full offline/online procedure, and the experiments next test whether this reversible representation preserves task performance better than standard baselines.

6 EXPERIMENTS

The experiments are organized around the paper’s main claims: conventional merging fails for low-rank models, RMM preserves more task performance, and its basis size exposes a storage–accuracy trade-off. We evaluate these claims across compression methods, ranks, tasks, and model architectures.

6.1 EXPERIMENTAL SETUP

The primary evaluation uses eight RoBERTa-base models, each fine-tuned for a different task (Zhuang et al., 2021). We compress each model with PT-SVD or LoRA at ranks $r \in \{16, 32, 64, 128\}$, then apply RMM to the resulting low-rank representations. The reviewer-facing comparisons are Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024), with individual compressed models before merging reported as the upper reference. Appendix A.2 broadens the evidence to OPT-1.3b (Zhang et al., 2022) and ViT-B/32 (Radford et al., 2021), including different task counts and a vision setting.

Metrics follow the GLUE benchmark protocol (Wang et al., 2018). Accuracy is used for QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification),

Algorithm 1 Reversible Model Merging (RMM)

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1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 
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MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment); Matthews correlation is used for CoLA, and Pearson correlation for STS-B. We report the average across tasks to summarize performance and pair it with relative storage, measured against storing all individual task-specific adapters. Single-merged-model baselines have fixed cost $\frac{r}{rn} = \frac{1}{n}$, while RMM with p basis components has cost $\frac{p(r+n)+r}{rn}$ as derived in Section 5.1.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 provide the main empirical evidence for the reversible-merging contribution. For both PT-SVD and LoRA, and for every rank $r \in \{16, 32, 64, 128\}$, RMM outperforms TA, TIES, and DARE by large margins. The low-rank regime makes the contrast clearest: at $r = 16$ and $r = 32$, the baselines fall to roughly 31–42% average score and are especially weak on tasks such as CoLA and STS-B. RMM with $p = 2$ already improves substantially, including 57.24% under PT-SVD and 51.41% under LoRA at $r = 16$, and RMM with $p = 3$ reaches 72.22% and 58.64% at the same rank. At $r = 64$ and $r = 128$, where the source compression is less severe, RMM remains ahead: with $p = 3$, it obtains 64.34% and 60.58% under PT-SVD, and 55.40% and 61.35% under LoRA, while single-model baselines remain near 40%. These results support the central claim that reconstructability mitigates the destructive interference that limits one-model merging.

The results also reveal a limitation of the contribution: RMM’s relative gain over baselines decreases as rank increases. A plausible explanation is that higher-rank task representations occupy richer and less aligned subspaces, making them harder to approximate with a small shared basis. Even so, the aggregate pattern is consistent across compression regimes and ranks: RMM exchanges additional offline storage for substantially better task performance. This cost is typically CPU-side memory rather than runtime model quality, so the method is most attractive when a modest storage increase is acceptable in return for preserving the specialized behavior of the original low-rank models.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: Main PT-SVD result: GLUE performance and storage for merging eight compressed RoBERTa-base models across ranks $r = 16, 32, 64, 128$.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: Main LoRA result: GLUE performance and storage for merging eight compressed RoBERTa-base models across ranks $r = 16, 32, 64, 128$.

6.3 SCALABILITY OF RMM

We finally test the storage-scaling claim as the number of models n increases. Figures 4 and 5 plot the storage ratio relative to storing all task-specific models for RMM with $p = 2$ and $p = 3$. The trend is favorable: as more tasks are added, the relative storage cost drops. This behavior is a practical contribution of the reversible formulation. Once the basis is stored, each additional task mainly adds a new row to the coefficient matrix, so storage grows sublinearly relative to keeping all adapters. Standard alternatives force a less attractive choice: store one merged model and lose task-specific performance, or store every task model and scale linearly. RMM therefore directly targets

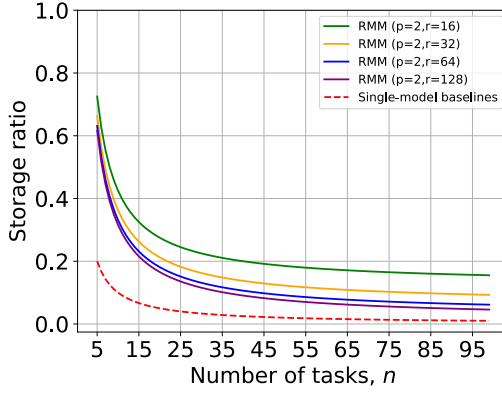


Figure 4: Scalability contribution for RMM with $p = 2$: relative storage decreases as the number of task models grows.

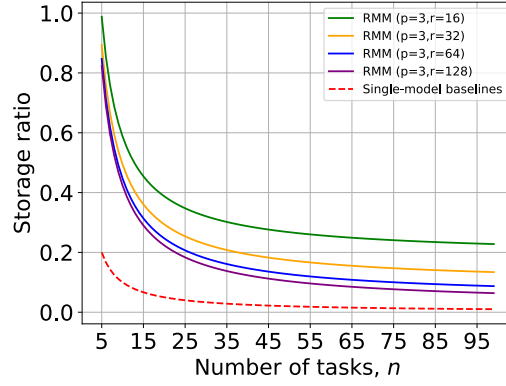


Figure 5: Scalability contribution for RMM with $p = 3$: relative storage remains below storing all task models and improves with more tasks.

large multi-task, continual, and federated settings where many low-rank models must be managed without discarding their task-specific accuracy.

7 CONCLUSION

This paper contributes a reversible alternative to standard low-rank model merging. First, we show that conventional methods can fail severely when applied to LoRA or PT-SVD weights. Second, we reformulate merging as storing a compact reconstruction basis rather than collapsing all task knowledge into one parameter vector. Third, we instantiate this formulation as RMM, a data-free method with a closed-form basis and task-specific coefficients. The result is a representation that stores substantially fewer components than all original models while remaining more faithful to task-specific behavior than a single aggregate. Its tunable basis size lets practitioners choose the storage–reconstruction point appropriate for their deployment constraints.

The broader message is that the contribution of merging need not be a single final model. For compressed task models, preserving the ability to reconstruct specialized adapters can be more valuable than forcing an aggressive one-shot merge. This perspective broadens the design space for efficient multi-task deployment and suggests future work on performance-aware compression, routing, and reconstruction strategies.

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A APPENDIX

A.1 PROOF OF THEOREM 1

The proof makes explicit why the RMM basis in Theorem 1 is the optimal reconstruction subspace. Let $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$. By the Frobenius norm identity, $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, where $\text{Tr}(\cdot)$ denotes the trace of a square matrix. Expanding the reconstruction error gives:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

The term inside the trace expands as follows:

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Taking the trace and using cyclicity together with $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ yields:

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

The first term, $\text{Tr}(\mathbf{X}^\top \mathbf{X})$, is fixed with respect to $\mathbf{W}$. Therefore, minimizing the RMM reconstruction error is equivalent to maximizing the retained projected variance, $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$:

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), the optimizer is obtained by setting the columns of $\mathbf{W}$ to the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. Thus the RMM basis is exactly the PCA reconstruction subspace: it maximizes projected variance and equivalently minimizes squared reconstruction error. If $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ is the SVD of $\mathbf{X}$, then $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$, so these eigenvectors are the right singular vectors, i.e., the columns of $\mathbf{V}$. The practical implementation can therefore use the top- p right singular vectors of $\mathbf{X}$, which is numerically stable and avoids explicitly forming $\mathbf{X}^\top \mathbf{X}$ when r is large.

A.2 ADDITIONAL RESULTS

This appendix extends the empirical contribution beyond the main RoBERTa-base setting. For **OPT-1.3b**, we evaluate six natural language understanding tasks from GLUE and related benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we test whether the same reversible-merging behavior appears in vision classification across MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: Additional language-model evidence: GLUE and BoolQ performance for merging six PT-SVD-compressed OPT-1.3b models across ranks $r = 16, 32, 64, 128$.

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Additional vision evidence: performance and storage for merging seven PT-SVD-compressed ViT-B/32 models across ranks $r = 16, 32, 64, 128$.

Table 3 shows that the OPT-1.3b results support the main claim: RMM with both $p = 2$ and $p = 3$ outperforms the baselines, and $p = 3$ moves closer to the individual compressed models while storing fewer parameters. Table 4 provides the same check for seven ViT-B/32 models. In the vision setting, RMM with $p = 3$ is less than 3% below the original models on average while using only 59% of the storage required to retain all models at $r = 128$. These additional experiments reinforce that the contribution is not tied to one architecture: reversible bases can preserve compressed-model behavior while reducing storage.

A.3 RESULTS FOR 5 TASKS

This subsection provides the tabular evidence behind the motivating observation in Section 4. We use RoBERTa-base as the pretrained model and evaluate merging baselines on five GLUE tasks: QNLI, MRPC, SST-2, MNLI, and QQP. Figures 1 and 2 in Section 4 summarize the same results visually. Table 5 corresponds to Figure 1 and reports the LoRA setting and the gains from RMM, while Table 6 corresponds to Figure 2 and reports the corresponding PT-SVD setting.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: Motivating LoRA table: GLUE performance for merging five compressed RoBERTa-base models across ranks $r = 16, 32, 64, 128$.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: Motivating PT-SVD table: GLUE performance for merging five compressed RoBERTa-base models across ranks $r = 16, 32, 64, 128$.